WIT-SOC

Wire-Interposition Trust Layer for Autonomous AI SOCs

System Architecture, Implementation, Security, Testing and Demonstration Guide

Report detail	Description
Project type	Academic Cybersecurity, Artificial Intelligence & Hardware Security Project (FF No. 180)
Institution	Vishwakarma Institute of Technology, Pune • Department of Computer Engineering (IoTCSBT)
Academic term	Academic Year 2026–27 • Semester 5 • Group No. 9
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System model	Multi-node autonomous wire-interposition security architecture with out-of-band hardware quarantine
Prepared	18 October 2026
Purpose	Team handover, academic jury defense, and shared understanding of the complete system

Based on project architecture and live multi-workstation verification results available on 18 October 2026.

1. Executive Summary

WIT-SOC (Wire-Interposition Trust Layer for Autonomous AI Security Operations Centers) is an intelligent cybersecurity system designed to eliminate the critical 2-to-5 second latency gap inherent in traditional reactive SOC pipelines (Attack → Collect Logs → AI Inference → Final Decision → Isolation). During this computational window, fast-moving threats such as automated ransomware, network worms, and credential-dumping exploits laterally compromise entire subnets before classical firewalls intervene. Furthermore, software-only endpoint agents can be bypassed via local OS privilege escalation.

WIT-SOC introduces two structural innovations: **Speculative Execution for Network Security** and an **Out-of-Band Hardware Mitigation Layer**. Early sub-threshold anomalies trigger temporary, reversible micro-containment at moderate confidence (≥ 50%), while high-confidence attacks (≥ 85%) trigger an independent hardware anchor (Raspberry Pi) that severs the compromised host's network interface out-of-band via UDP kill packets.

Live checks verified dual-layer AI telemetry ingestion, sub-second threat scoring, speculative Yellow state transitions, permanent Red lockout, real-time Streamlit dashboard visualization, persistent SQLite SIEM audit logging, and automated Wi-Fi interface disconnection across physical network workstations.

2. Project Goals

- Demonstrate speculative containment concepts using a multi-node workstation and gateway topology.
- Eliminate the 2–5 second threat propagation window through fast-path heuristics and microsecond ML inference.
- Provide autonomous, out-of-band hardware enforcement independent of host OS software integrity.
- Implement reversible containment tiers to prevent operational business disruption from false alarms.
- Classify real-world network intrusions (benign, reconnaissance, and volumetric DDoS/ransomware) using CIC-IDS2017 features.
- Persist audit ledgers for compliance and forensic analysis in an ACID-compliant SQLite SIEM database.
- Expose a real-time web monitoring console displaying live gauges, telemetry charts, and emergency reset controls.

3. High-Level Architecture

The system consists of workstation endpoint agents, an asynchronous FastAPI gateway running a dual-layer threat engine, a trained Random Forest classifier, an out-of-band hardware controller (Raspberry Pi / Mock Pi), an SQLite SIEM ledger, and a Streamlit SOC dashboard.

Component	Responsibility	Usage
Endpoint Agent Daemon	Workstation enforcement listener	Binds to UDP :9999; executes native OS network disconnection upon receiving hardware kill commands.
Attack & Telemetry Simulator	Traffic generation & decoy vector	Streams 5-feature network flow metrics mimicking clean background usage or 3-stage intrusion attacks.
Central Gateway (FastAPI)	Ingestion, state management & API	Extracts client IP, routes telemetry through dual-layer AI, manages Green/Yellow/Red system state.
Random Forest AI Model	Multi-class threat confidence scoring	Evaluates tabular flow vectors; outputs exact class probabilities across Benign, Suspicious, and Malicious.
Hardware Trust Layer	Out-of-band verification & actuation	Polls gateway every 0.5s; actuates physical breadboard LEDs; dispatches UDP isolation datagrams.
SIEM Audit Database	Persistent transactional logging	Stores timestamped records of every flow, threat score, verdict, detection engine, and quarantined IP.
Streamlit SOC Dashboard	Analyst visualization & control	Displays live threat gauges, traffic stream line charts, policy breakdown, and 1-click system reset.

4. Runtime Topology and Ports

Service	Host port(s)	Role & Protocol
FastAPI Gateway	8000	HTTP REST API; ingests `/api/v1/telemetry` and exposes `/api/v1/status`.
Streamlit Dashboard	8501	Web UI; browser interface for real-time threat tracking and emergency system reset.
Endpoint Agent	9999	UDP socket listener; receives out-of-band `EXECUTE_ISOLATION` kill commands.
Mock Hardware Controller	Internal polling	0.5s polling client over HTTP; out-of-band UDP datagram sender to port 9999.
Raspberry Pi Controller	GPIO 17, 27, 22	Physical breadboard hardware; actuates Red, Yellow, and Green LEDs; sends UDP kill packets.

5. Technology Stack

Area	Technology	Purpose
Frontend / UI	Streamlit 1.35+, Altair, HTML5/CSS3	Interactive real-time SOC monitoring dashboard and analyst console.
Backend / API	Python 3.11+, FastAPI, Uvicorn, ASGI	High-concurrency asynchronous API server and core state manager.
AI / Threat Engine	Scikit-Learn, Pandas, NumPy, Joblib	Random Forest Classifier trained on CIC-IDS2017 intrusion telemetry.
Audit Persistence	SQLite3 (ACID compliant)	Hardware-verified event logs and historical telemetry ledger.
Hardware Layer	Raspberry Pi (BCM GPIO), Gpiozero	Physical LED status signaling and zero-trust out-of-band enforcement.
Endpoint Severance	Python Socket, Windows Netsh, Linux Nmcli	Hardware-commanded network interface disconnection.
Automation	Windows Batchfiles (.bat)	1-click deployment launchers for all master and endpoint components.

6. How the System Works

6.1 Telemetry Collection & Feature Extraction

Workstation endpoints continuously measure network flow behavior and stream 5 standardized telemetry metrics: `flow_duration` (ms), `flow_bytes_sec` (rate), `total_fwd_packets` (count), `total_bwd_packets` (count), and `fwd_packet_length_mean` (size).

6.2 Dual-Layer AI Detection Engine

Upon receiving telemetry, the Central Gateway evaluates threat confidence using a two-stage pipeline:

- **Layer 1 (Fast-Path Heuristics, < 10ms):** Evaluates volumetric flood triggers (`flow_bytes_sec > 1,000,000` or `total_fwd_packets > 1,000`). High-volume DDoS/ransomware attacks are detected immediately.
- **Layer 2 (Random Forest Classifier):** If sub-threshold, the 5 features are fed to the pre-compiled ML model, calculating exact probabilities across Benign, Suspicious, and Malicious classes.

6.3 Speculative Execution Policy Matrix

Threat Score	Action	Signal	Hardware Output	System State & Behavior
< 50.0%	`ALLOW`	**GREEN**	Green LED ON	Nominal State. Standard employee traffic permitted without restriction.
50.0% – 84.9%	`TEMP_ISOLATE`	**YELLOW**	Yellow LED ON	**Speculative Micro-Containment:** Sub-threshold anomaly flagged. Monitored; fully reversible if traffic normalizes.
≥ 85.0%	`PERM_LOCK`	**RED**	Red LED ON + UDP Kill	**Quarantine Enforced:** Out-of-band UDP kill packet fired to port 9999. Host Wi-Fi adapter severed instantly.

6.4 Out-of-Band Hardware Quarantine

When a critical threat is confirmed (≥ 85%), the hardware controller actuates the Red LED and fires a datagram containing `EXECUTE_ISOLATION` over UDP to port 9999 of the attacker's extracted IP. The workstation daemon executes `netsh wlan disconnect` (or `nmcli radio wifi off`), severing the radio layer in under a second.

6.5 Persistent SIEM Audit Logging

Every incoming telemetry event is committed to `siem_audit.db`. The schema stores timestamp, client IP, raw flow metrics, threat score, verdict, hardware signal, and the detection engine used (Heuristic or Random Forest).

7. API Reference

Method	Endpoint	Purpose
POST	`/api/v1/telemetry`	Ingests workstation flow metrics, extracts client IP, evaluates AI threat score, updates state.
GET	`/api/v1/status`	Polled every 0.5s by hardware controllers; returns hardware signal, action, score, and isolated IP.
GET	`/api/v1/logs`	Returns recent SIEM audit records from SQLite for the monitoring dashboard.
GET	`/api/v1/stats`	Returns aggregated packet counts, verdict distributions, and list of all isolated IP addresses.
POST	`/api/v1/reset`	Clears active network quarantine, unfreezes state, and resets hardware signals back to GREEN.

8. Security and Zero-Trust Integrity

- **Out-of-Band Integrity:** Security enforcement is decoupled from the host OS. Even if malware achieves local root/SYSTEM privileges on the workstation, it cannot alter the independent hardware trust anchor.
- **Reversible Speculative Safety:** Temporary isolation during Yellow states automatically subsides if traffic normalizes, eliminating false-positive business interruption.
- **Hardware Fail-Safe:** If the hardware controller loses network connectivity to the gateway, all LEDs default to Yellow as an explicit visual alert.
- **Zero Hardcoded Secrets:** Network communication relies on dynamic runtime IP extraction and standard transport protocols.

9. Live Verification Results

These checks were performed interactively on 18 October 2026 across two physical laptops on a local Wi-Fi subnet (PC 1: `172.20.70.36` | PC 2: `172.20.70.210`).

Test	Observed result	Status / scope
Gateway Startup	FastAPI gateway initialized on `0.0.0.0:8000`; model loaded successfully.	**Pass**
Dashboard Launch	Streamlit dashboard initialized at `http://localhost:8501`; status nominal GREEN.	**Pass**
Virtual Hardware Init	Mock Pi controller initialized; polling 0.5s loop; GREEN LED active.	**Pass**
Agent Initialization	Endpoint severing agent bound to UDP :9999 on PC 2; listening for kill packets.	**Pass**
Clean Flow Telemetry	Packet #1 processed; threat score 0.0%; verdict ALLOW; hardware remains GREEN.	**Pass**
Speculative Anomaly	Packet #2 (port scan) processed; threat score 75.0%; hardware switched to YELLOW.	**Pass (Speculative)**
Volumetric Attack Flood	Packet #3 (DDoS/exfiltration) processed; threat score 100.0%; verdict PERM_LOCK.	**Pass (Critical)**
Dynamic IP Extraction	Gateway automatically identified attacker IP as `172.20.70.210` without manual config.	**Pass**
Out-of-Band UDP Dispatch	Hardware layer dispatched `EXECUTE_ISOLATION` datagram to `172.20.70.210:9999`.	**Pass**
Hardware Wi-Fi Severance	PC 2 agent received kill command; dropped Wi-Fi card via native Netsh command.	**Pass (< 1s TTC)**
Emergency System Reset	Dashboard reset button cleared lock; returned gateway and hardware state to GREEN.	**Pass**
Second Demo Verification	PC 2 reconnected and rerun; system demonstrated 100% reproducible quarantine.	**Pass (Repeatable)**

10. Test Run Evidence

Field	Value for Live Test Run
Master Gateway Node	`172.20.70.36` (FastAPI :8000, Streamlit :8501, Mock Pi Controller)
Target Workstation Node	`172.20.70.210` (Severing Agent :9999, Telemetry Simulator)
Malicious Flow Profile	Duration: 5.0ms • Rate: 5,000,000 bytes/sec • Forward Packets: 2500 • Packet Length: 6000
AI Model Confidence	100.0% Malicious (Random Forest Classifier + Volumetric Fast-Path Heuristic)
Hardware Action Taken	State: `PERM_LOCK` • Signal: `RED LED` • Datagram: `EXECUTE_ISOLATION`
Endpoint Agent Execution	`Disconnection request was completed successfully for interface "Wi-Fi".`

11. Automated Assurance & Regression Suite

An automated integration test suite (`tests/test_full_suite.py`) executes **17 comprehensive assertions** covering model loading, multi-class predictions, database CRUD operations, policy transitions, UDP transmission, and dashboard syntax. All 17 tests passed with zero failures in < 3 seconds.

12. Limitations and Claims to Avoid

- The current endpoint agent runs in user/system space utilizing OS-level netsh/nmcli utilities. It demonstrates the out-of-band wire-interposition principle but is not yet implemented as an in-kernel eBPF packet drop driver.
- The physical Raspberry Pi hardware module is validated via its software controller and verified virtual controller; physical breadboard LEDs will be demonstrated during final on-campus hardware reviews.
- Do not claim the model detects arbitrary zero-day binaries; the model is specifically trained on network flow telemetry features (duration, rates, packet counts).
- Do not claim enterprise data center deployment; position the Raspberry Pi implementation as an educational prototype mirroring modern SmartNIC/DPU wire-interposition architectures.

13. Demo Runbook

Project Directory: `C:\Users\Chira\.gemini\antigravity\scratch\WIT-SOC`

- **Step 1 (Master Setup):** Connect both laptops to same Wi-Fi/Hotspot. Run `ipconfig` on PC 1 to get IPv4 (`172.20.70.36`).
- **Step 2 (Start Backend):** Run `run_backend.bat` . Verify Uvicorn starts on `0.0.0.0:8000` and AI model loads.
- **Step 3 (Start Dashboard):** Run `run_dashboard.bat` . Open `http://localhost:8501` ; verify status is GREEN.
- **Step 4 (Start Hardware Controller):** Run `run_mock_pi.bat` . Press Enter for localhost. Verify GREEN LED is active.
- **Step 5 (Start Endpoint Agent on PC 2):** On PC 2, run `run_endpoint_agent.bat` . Verify UDP socket listens on port 9999.
- **Step 6 (Trigger Attack on PC 2):** Run `python telemetry_simulator.py --attack` . Enter PC 1's IP.
- **Step 7 (Observe Mitigation):** Show Stage 1 (Green), Stage 2 (Yellow warning), Stage 3 (Red Perm Lock). Show PC 2 Wi-Fi dropping.
- **Step 8 (System Reset):** Click "Emergency System Reset" in dashboard sidebar to return to Green. Toggle PC 2 Wi-Fi to reconnect.

14. Suggested Team Ownership

Area	What the teammate should understand / own during jury evaluation
AI / Machine Learning	Feature extraction, Random Forest hyperparameters, CIC-IDS2017 dataset, multi-class mapping, fast-path heuristics.
Backend & API Gateway	FastAPI async routes, dynamic client IP extraction, state transitions, CORS, SQLite SIEM database schema.
Hardware & OOB Trust	Raspberry Pi GPIO pinouts, breadboard resistors, out-of-band UDP socket dispatching, zero-trust hardware principles.
Endpoint & Testing	Endpoint agent daemon, Netsh/Nmcli interface dropping, 3-stage attack simulation, multi-laptop empirical verification.

15. Conclusion

Live verification demonstrates an operational, end-to-end autonomous cyber defense system. By combining **Speculative Micro-Containment** with an **Out-of-Band Hardware Mitigation Layer**, WIT-SOC eliminates the dangerous 2-to-5 second latency window exploited by lateral-moving ransomware. The multi-laptop deployment successfully proved sub-second detection, accurate dynamic IP extraction, real-time visual telemetry tracking, and reliable automated network isolation. The prototype successfully validates the wire-interposition security paradigm and provides a solid foundation for future offloading to enterprise SmartNICs and DPUs.

Appendix A — Useful Commands & Endpoints

Purpose	Command / HTTP Endpoint
Check Gateway Status	`curl.exe http://localhost:8000/api/v1/status`
View SIEM Audit Statistics	`curl.exe http://localhost:8000/api/v1/stats`
Trigger System Reset	`curl.exe -X POST http://localhost:8000/api/v1/reset`
Run Automated Test Suite	`.venv\Scripts\python.exe tests\test_full_suite.py`
Query SQLite SIEM Database	`sqlite3 backend\siem_audit.db "SELECT * FROM telemetry_logs ORDER BY id DESC LIMIT 5;"`
Re-enable Wi-Fi on Windows	`netsh interface set interface "Wi-Fi" enable`

Appendix B — Raspberry Pi Physical Hardware Pinout

Component	Color	Raspberry Pi Pin (BCM)	Physical Header Pin	Wiring & Connection
Red LED	Red	**GPIO 17**	Physical Pin 11	Anode → Pin 11 • Cathode → 220Ω → GND (Pin 6/9)
Yellow LED	Yellow	**GPIO 27**	Physical Pin 13	Anode → Pin 13 • Cathode → 220Ω → GND (Pin 6/9)
Green LED	Green	**GPIO 22**	Physical Pin 15	Anode → Pin 15 • Cathode → 220Ω → GND (Pin 6/9)